

Chapter 6: Symmetries

Quantum Mechanics

1 Introduction to Symmetry

Definition 1.1 (Symmetry). [def:symmetry] A *symmetry* of a system is some transformation, such that the system is invariant before and after the transformation.

Definition 1.2 (Symmetry of Quantum System). [def:symmetry-qm] A *symmetry* of a quantum system is a transformation that leaves the Hamiltonian invariant.

Definition 1.3 (Types of Symmetry). [def:symmetry-types] There are several types of symmetries in quantum mechanics, including:

- *Translation symmetry*: The system is invariant under some spatial translations.
- *Rotation symmetry*: The system is invariant under some rotations.
- *Reflection symmetry*: The system is invariant under reflections on some axis/planes.
- *Inversion symmetry*: The system is invariant under inversion of all coordinates.
- *Time-reversal symmetry*: The system is invariant under the reversal of time.

Theorem 1.1 (Symmetry Condition on the Hamiltonian). [thm:symmetry-condition-hamiltonian] $\hat{U}$ is a definition 1.2 of $\hat{H}$ iff

$$\hat{U}^\dagger \hat{H} \hat{U} = \hat{H}.$$

Corollary 1.1 (Commutator Form). [cor:commutator-form-symmetry] For unitary $\hat{U}$,

$$\hat{U}^\dagger \hat{H} \hat{U} = \hat{H} \iff \text{comm}(\hat{U}, \hat{H}) = 0.$$

Proof. Left-multiply by $\hat{U}$ and use $\hat{U}\hat{U}^\dagger = \hat{I}$:

$$\hat{U}^\dagger \hat{H} \hat{U} = \hat{H} \iff \hat{H} \hat{U} = \hat{U} \hat{H} \iff \text{comm}(\hat{U}, \hat{H}) = 0.$$

□

Remark 1.1 (“ $\hat{A}$ Obtained After Applying $\hat{U}$ ”). [rem:conjugation-vocabulary] The phrase “ $\hat{A}$ obtained after applying $\hat{U}$ ” refers to the conjugation $\hat{U}^\dagger \hat{A} \hat{U}$, since this preserves the expectation value.

2 Groups

Definition 2.1 (Group). [def:group] A *group* consists of the following data and assumptions:

- A set G .
- A binary operation $\cdot : G \times G \rightarrow G$ (the codomain being G bakes in closure).
- An identity element $e \in G$.
- An inverse function $(\cdot)^{-1} : G \rightarrow G$, written $g \mapsto g^{-1}$.
- *Associativity*: $(g_1 \cdot g_2) \cdot g_3 = g_1 \cdot (g_2 \cdot g_3)$ for all $g_1, g_2, g_3 \in G$.
- *Identity axiom*: $e \cdot g = g \cdot e = g$ for all $g \in G$.
- *Inverse axiom*: $g \cdot g^{-1} = g^{-1} \cdot g = e$ for all $g \in G$.

A collection with a well-behaved multiplication and inverses.

Example 2.1 (Real Numbers Under Addition). [ex:reals-group] Choose:

- Set $G := \mathbb{R}$ (the real numbers).
- Operation $\cdot := +$ (ordinary addition).
- Identity $e := 0$.
- Inverse $a^{-1} := -a$.

Associativity, identity, and inverse axioms follow from the usual properties of addition.

Example 2.2 (Boolean Set Under XOR). [ex:boolean-group] Choose:

- Set $G := \{\text{tt}, \text{ff}\}$ (equivalently $\{0, 1\}$).
- Operation $\cdot := \oplus$ (XOR, equivalent to addition mod 2).
- Identity $e := \text{ff}$ (equivalently 0).
- Inverse $g^{-1} := g$ (each element is self-inverse).

All axioms hold by direct verification.

Definition 2.2 (Continuous Group). [def:continuous-group] A group is *continuous* (a *Lie group*) if its elements are parametrized by a [continuous parameter space](#), and multiplication and inversion vary smoothly with the parameters. The real numbers under addition is an example.

Definition 2.3 (Discrete Group). [def:discrete-group] A group is *discrete* if its elements form a [discrete set](#) (finite or [countable](#)), with no continuous parameter connecting distinct elements. The Boolean set under XOR is an example.

Definition 2.4 (Continuous Symmetry). [def:continuous-symmetry]

Definition 2.5 (Discrete Symmetry). [def:discrete-symmetry]

3 Translation Operator

Definition 3.1 (Translation Operator). [def:translation-operator] Intuitively, a *translation operator* $\hat{T}(\vec{r}_0)$ (translates a state by a vector $\vec{r}_0$) acts on a state $|\psi\rangle$ should let $|\psi\rangle$ be translated by a vector $\vec{r}$, i.e.

$$\hat{T}(\vec{r}_0) |\psi\rangle = |\psi(\vec{r} - \vec{r}_0)\rangle$$

Definition 3.2 (Translation Operator (1D)). [def:translation-operator-1d] In one dimension, the translation operator $\hat{T}(a)$ shifts a state by a real distance $a \in \mathbb{R}$. Its action on a position-space wavefunction is

$$\hat{T}(a) \psi(x) = \psi(x - a).$$

Theorem 3.1 (Translation Operators Form a Symmetry Group). [thm:translation-symmetry-group] Suppose $\hat{T}(a)$ is a symmetry operator of a Hamiltonian $\hat{H}$ for every $a \in \mathbb{R}$. Then the collection of translation operators forms a group under operator composition:

- Set $G := \{\hat{T}(a) : a \in \mathbb{R}\}$.
- Operation $\cdot$: operator composition, $\hat{T}(a) \cdot \hat{T}(b) = \hat{T}(a + b)$.
- Identity $e := \hat{T}(0) = \hat{I}$.
- Inverse $\hat{T}(a)^{-1} := \hat{T}(-a)$.

All axioms follow from addition on $\mathbb{R}$.

Proof.

□

Theorem 3.2 (Exponential Form of the Translation Operator). [thm:translation-exponential] The one-dimensional translation operator can be written as:

$$\hat{T}(a) = e^{-ia\hat{p}/\hbar}$$

Proof. Expand the exponential in a Taylor series at 0

$$\hat{T}(a) = e^{-ia\hat{p}/\hbar} = \sum_{n=0}^{\infty} \frac{(-ia/\hbar)^n}{n!} \cdot \hat{p}^n = \sum_{n=0}^{\infty} \frac{(-ia\hat{p})^n}{n!\hbar^n}$$

According to the definition of the momentum operator, $\hat{p} = -i\hbar \frac{d}{dx}$, so we can substitute this into the series expansion:

$$\hat{T}(a) = \sum_{n=0}^{\infty} \frac{(-ia(-i\hbar \frac{d}{dx}))^n}{n!\hbar^n} = \sum_{n=0}^{\infty} \frac{(-a(\frac{d}{dx}))^n}{n!}$$

So,

$$\hat{T}(a)\psi(x) = \sum_{n=0}^{\infty} \frac{(-a(\frac{d}{dx}))^n}{n!} \psi(x) = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}(x)$$

Then, we do a Taylor expansion of $\psi(x)$ centered at $x + a$:

$$\psi(x) = \sum_{n=0}^{\infty} \frac{\psi^{(n)}(x+a)}{n!} (x - (x+a))^n = \sum_{n=0}^{\infty} \frac{\psi^{(n)}(x+a)}{n!} (-a)^n = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}(x+a)$$

Substituting $x \rightarrow x - a$ in the previous identity,

$$\psi(x-a) = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}((x-a)+a) = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}(x).$$

Which means,

$$\hat{T}(a)\psi(x) = \psi(x-a)$$

□

Theorem 3.3 ($\hat{T}(a)$ is a Unitary Operator). [thm:translation-unitary] *The translation operator $\hat{T}(a)$ is a unitary operator, i.e. $\hat{T}^\dagger(a) = \hat{T}^{-1}(a) = \hat{T}(-a)$.*

Remark 3.1 (Intuition about). [rem:intuition-about]

Proof. **TODO:** Proof to be written after clarifying preferred approach with the lecturer — direct from the action definition (definition 3.2), or via the exponential form (theorem 3.2). Email sent September 23, 2026. □

Theorem 3.4 (Position Operator After Applying Translation). [thm:position-under-translation] *Applying $\hat{T}(a)$ to the position operator $\hat{x}$ (see remark 1.1) gives*

$$\hat{T}(a)^\dagger \hat{x} \hat{T}(a) = \hat{x} + a\hat{I}.$$

Proof. Define $\hat{x}' := \hat{T}(a)^\dagger \hat{x} \hat{T}(a)$, and consider its action on a wavefunction $\psi(x)$:

$$\hat{x}'\psi(x) = \hat{T}(a)^\dagger \hat{x} \hat{T}(a) \psi(x)$$

Since operators are associative, we have

$$\hat{x}'\psi(x) = \hat{T}(a)^\dagger \hat{x} (\hat{T}(a) \psi(x)) = \hat{T}(a)^\dagger (\hat{x} \psi(x-a)) = \hat{T}(a)^\dagger (x \psi(x-a))$$

Treat $g(x) = \hat{x} \psi(x-a)$, so, $\hat{T}(a) g(x) = g(x) = x \hat{T}(a) \psi((x-a)-a)$.

Notice that $\hat{T}(a)$ is a unitary operator, ie $\hat{T}(a)^\dagger = \hat{T}(a)^{-1} = \hat{T}(-a)$, we have

$$\hat{x}'\psi(x) = \hat{T}(-a) (x \psi(x-a)) = (x+a) \psi((x+a)-a) = (x+a) \psi(x)$$

Rewrite the right-hand side in operator form:

$$(x+a) \psi(x) = x \psi(x) + a \psi(x) = \hat{x} \psi(x) + a\hat{I} \psi(x) = (\hat{x} + a\hat{I}) \psi(x).$$

So $\hat{x}' \psi = (\hat{x} + a\hat{I}) \psi$ for every ψ . Since two operators agreeing on all inputs are equal, we conclude

$$\hat{x}' = \hat{x} + a\hat{I}.$$

□

Theorem 3.5 (Momentum Operator After Applying Translation). [thm:momentum-under-translation] Applying $\hat{T}(a)$ to the momentum operator $\hat{p}$ (see remark 1.1) gives

$$\hat{T}(a)^\dagger \hat{p} \hat{T}(a) = \hat{p}.$$

Proof. Define $\hat{p}' := \hat{T}(a)^\dagger \hat{p} \hat{T}(a)$, and consider its action on a wavefunction $\psi(x)$:

$$\hat{p}'\psi(x) = \hat{T}(a)^\dagger \hat{p} \hat{T}(a) \psi(x) = \hat{T}(a)^\dagger \hat{p} \psi(x - a)$$

According to the definition of definition momentum operator position, $\hat{p} = -i\hbar d/dx$, plug in:

$$\hat{p}'\psi(x) = -\hat{T}(a)^\dagger i\hbar \frac{d\psi(x - a)}{dx}$$

Use $\hat{T}(a)^\dagger = \hat{T}(-a)$,

$$\hat{p}'\psi(x) = -\hat{T}(-a) i\hbar \frac{d\psi(x - a)}{dx} = -i\hbar \frac{d\psi(x)}{dx} = \hat{p}\psi(x)$$

So $\hat{p}'\psi = \hat{p}\psi$ for every ψ . Since two operators agreeing on all inputs are equal, we conclude

$$\hat{p}' = \hat{p}.$$

□

Theorem 3.6. For all operator, which only consists of $\hat{x}$ and $\hat{p}$, the conjugation action of $\hat{T}(a)$ on the operator is equivalent to replacing every $\hat{x}$ with $\hat{x} + a\hat{I}$ and leaving every $\hat{p}$ unchanged. ie,

$$Q(\hat{x}, \hat{p}) \longmapsto \hat{T}(a)^\dagger Q(\hat{x}, \hat{p}) \hat{T}(a) = Q(\hat{x} + a\hat{I}, \hat{p})$$

Theorem 3.7 (Translation Symmetry Condition on the Potential). [thm:translation-condition-potential] Let $\hat{H}$ be a 1D Hamiltonian. If $\hat{T}(a)$ is a symmetry operator of $\hat{H}$, then the potential is invariant under translation by a :

$$V(x + a) = V(x) \quad \text{for all } x.$$

Proof. Since $\hat{T}(a)$ is a unitary symmetry, corollary 1.1 gives

$$\hat{T}(a)^\dagger \hat{H} \hat{T}(a) = \hat{H} \iff \text{comm}(\hat{H}, \hat{T}(a)) = 0.$$

Notice that $\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$, and by theorem 3.5 and theorem 3.4, we have

$$\hat{T}(a)^\dagger \hat{H} \hat{T}(a) = \hat{T}(a)^\dagger \left(\frac{\hat{p}^2}{2m} + V(\hat{x}) \right) \hat{T}(a) = \frac{\hat{p}^2}{2m} + V(\hat{x} + a\hat{I}).$$

Which means, $\hat{T}(a)$ is a symmetry of $\hat{H}$ iff

$$V(x) = V(x + a)$$

□

Theorem 3.8 (Continuous Translation Symmetry Forces Constant V). [thm:continuous-translation-constant-v] If $\hat{T}(a)$ is a symmetry of $\hat{H}$ for every $a \in \mathbb{R}$, then V is constant.

Theorem 3.9 (Bloch's Theorem). [thm:bloch-theorem] If $\hat{T}(a)$ is a symmetry of $\hat{H}$ for some $a \in \mathbb{R}$, stationary states of $\hat{H}$ have the form

$$\psi(x) = e^{iqx} u(x)$$

Where

- q is called the crystal momentum
- $u(x)$ is periodic and have a period of a

Remark 3.2 (Stationary States may not be Periodic). [rem:stationary-states-may-not-be-periodic] From theorem 3.9, we can see that the stationary states $\psi(x)$ is not periodic. However $u(x)$ is periodic with period a .

Proof. Since $\hat{T}(a)$ is a symmetry of $\hat{H}$, we have $\text{comm}(\hat{H}, \hat{T}(a)) = 0$. According to theorem commuting share eigenvectors,

$$\exists \lambda, \hat{H} |\psi\rangle = E |\psi\rangle \iff \exists \lambda, \hat{T}(a) |\psi\rangle = \lambda |\psi\rangle$$

Also, we may have

$$\begin{aligned} \langle \hat{T}(a) \psi | \hat{T}(a) \psi \rangle &= \langle \psi | \hat{T}(a)^\dagger \hat{T}(a) | \psi \rangle = \langle \psi | \psi \rangle \\ \langle \hat{T}(a) \psi | \hat{T}(a) \psi \rangle &= \langle \lambda \psi | \lambda \psi \rangle = \lambda^* \lambda \langle \psi | \psi \rangle \end{aligned}$$

This means that,

$$\lambda^* \lambda = 1 \implies \lambda = e^{i\theta} \text{ for some } \theta \in [0, 2\pi)$$

So, since $T(a)$ may be written as $e^{-iap/\hbar}$, we can argue that $a \propto \theta$. Where $q = -p/\hbar$ so that $\theta = qa$.

We can rewrite it into $\psi(x) = e^{iqx} u(x)$

Then, we have

$$\hat{T}(a) |\psi\rangle = e^{iqa} |\psi\rangle \implies \psi(x-a) = e^{iqa} \psi(x)$$

□

4 Rotational Symmetry

5 Parity/Inverse Symmetry

Example 5.1 (Parity Group). [ex:parity-group] Choose:

- Set $G := \{\hat{I}, \hat{P}\}$.
- Operation $\cdot$: operator composition.
- Identity $e := \hat{I}$.
- Inverse $\hat{P}^{-1} := \hat{P}$ (since $\hat{P}^2 = \hat{I}$, parity is self-inverse).

A discrete group with two elements.

6 Time-Reversal Symmetry